Psychiatry as Psychology with Medicine

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Abstract

This paper formalizes the thesis that psychiatry may be understood as psychology with medicine: a discipline that preserves the psychological object of inquiry—mind, behavior, cognition, affect, agency, meaning, and social relation—while adding medical diagnosis, somatic causality, pharmacology, risk triage, public health infrastructure, and institutional authority. The paper develops a mathematical model of psychiatric decision making, an economic and financial model of intervention portfolios, a political and geopolitical model of mental health systems, a warfare-economics model of combat stress and force readiness, and a statistical and machine learning framework for prediction, causal inference, and policy evaluation. The central result is not that psychiatry is reducible to either medicine or psychology, but that psychiatry is a hybrid control system whose distinctive power comes from jointly optimizing psychological and medical interventions under biological, social, institutional, ethical, and economic constraints.

The paper ends with "The End"

Keywords: psychiatry, psychology, medicine, biopsychosocial model, causal inference, pharmacology, mental health economics, military psychiatry, machine learning, public policy.

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1 Introduction

Psychiatry is often described as a branch of medicine concerned with mental disorders. Psychology is often described as the scientific study of mind and behavior. The thesis of this paper is more structural:

> *Psychiatry is psychology equipped with medical instruments, medical risk authority, somatic explanatory variables, pharmacological interventions, and institutional duties of care.*

This statement is not a slogan of reductionism. It does not say that mental distress is merely brain chemistry, nor that medication exhausts psychiatric care. Rather, it says that the psychiatric object is psychological in appearance and medical in jurisdiction. The psychiatrist confronts beliefs, affects, desires, memories, perceptions, impulses, trauma, relational patterns, and behavior, but must also assess sleep, metabolism, pain, infection, endocrine disturbance, neurodevelopment, neurodegeneration, substance exposure, drug interaction, suicide risk, violence risk, pregnancy, age, disability, and medical comorbidity.

Thus psychiatry differs from psychology not by abandoning psychology, but by adding medicine. It differs from general medicine not by abandoning the body, but by treating the body through the mind and the mind through the body.

2 Conceptual Architecture

2.1 The Basic Distinction

Let $\mathcal{P}$ denote the domain of psychological variables and $\mathcal{M}$ the domain of medical variables. A purely psychological model may be written as

$$Y = f(X_{\mathcal{P}}, \varepsilon),$$

where Y is a mental-health outcome and $X_{\mathcal{P}}$ includes cognition, affect, personality, behavior, trauma, attachment, habit, and social learning.

A psychiatric model extends this to

$$Y = f(X_{\mathcal{P}}, X_{\mathcal{M}}, A_{\mathcal{P}}, A_{\mathcal{M}}, I, \varepsilon),$$

where $X_{\mathcal{M}}$ includes biological and medical covariates, $A_{\mathcal{P}}$ denotes psychological interventions, $A_{\mathcal{M}}$ denotes medical interventions, and I denotes institutional and legal context.

Definition 1 (Psychiatric Extension)**.** *A psychiatric extension of a psychological model is a model in which the space of explanatory variables, interventions, risks, and institutional duties is enlarged from*

$$(\mathcal{P}, A_{\mathcal{P}})$$

to

$$(\mathcal{P} \times \mathcal{M}, A_{\mathcal{P}} \times A_{\mathcal{M}}).$$

2.2 A Visual Model

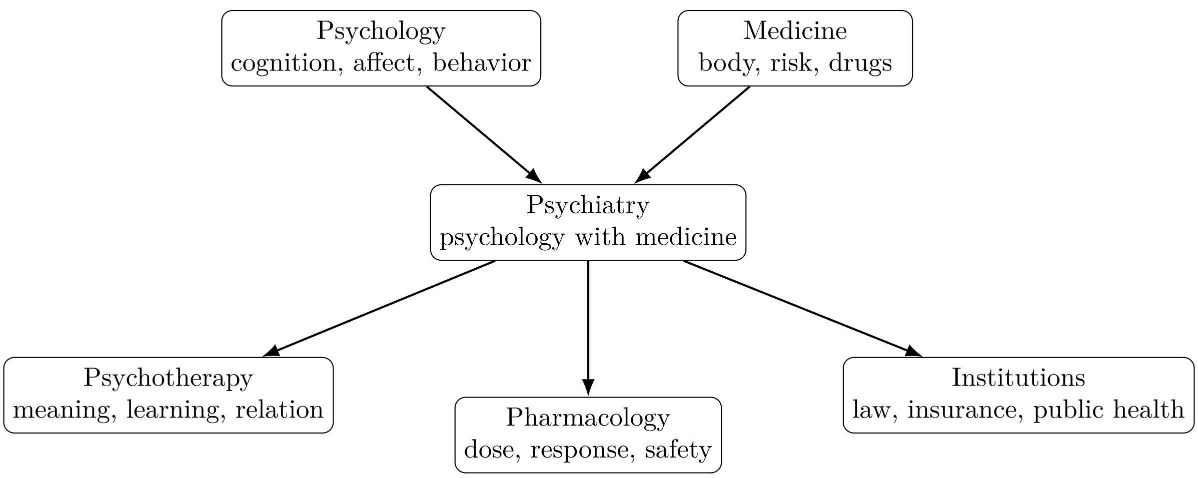

Figure 1: Psychiatry as the intersection of psychological science and medical authority.

3 Mathematical Formalization

3.1 State Space

Let a patient at time t be represented by a state vector

$$S_t = \begin{pmatrix} C_t \\ A_t \\ B_t \\ R_t \\ H_t \\ E_t \\ L_t \end{pmatrix},$$

where C_t denotes cognition, A_t affect, B_t behavior, R_t relational context, H_t health and somatic status, E_t economic environment, and L_t legal-institutional environment.

The psychiatrist chooses an action

$$u_t = (u_t^{\mathcal{P}}, u_t^{\mathcal{M}}, u_t^{\mathcal{S}}),$$

where $u_t^{\mathcal{P}}$ is a psychological intervention, $u_t^{\mathcal{M}}$ a medical intervention, and $u_t^{\mathcal{S}}$ a social or systems intervention.

The transition equation is

$$S_{t+1} = F(S_t, u_t, \eta_t),$$

where η_t captures uncertainty.

3.2 Objective Function

Let $Q(S_t)$ be quality of life, $D(S_t)$ distress, $K(u_t)$ treatment cost, and $R(S_t, u_t)$ risk. A general psychiatric objective is

$$\max_{\{u_t\}_{t=0}^T} \mathbb{E}\left[\sum_{t=0}^{T} \delta^t \left\{Q(S_t) - \lambda_D D(S_t) - \lambda_K K(u_t) - \lambda_R R(S_t, u_t)\right\}\right],$$

subject to clinical, ethical, legal, and resource constraints.

3.3 Psychology as a Boundary Case

Proposition 1 (Psychology as the Nonmedical Boundary of Psychiatry)**.** *Suppose the medical action set $A_{\mathcal{M}}$ contains only a null action $\varnothing$, and suppose all medical covariates are either absent or clinically irrelevant. Then the psychiatric optimization problem reduces to a psychological intervention problem.*

Proof. The psychiatric action set is

$$A = A_{\mathcal{P}} \times A_{\mathcal{M}}.$$

If $A_{\mathcal{M}} = \{\varnothing\}$, then every admissible action has the form

$$u = (u^{\mathcal{P}}, \varnothing).$$

If medical covariates are absent or clinically irrelevant, then

$$F(S_t, u_t, \eta_t) = F_{\mathcal{P}}(S_t, u_t^{\mathcal{P}}, \eta_t).$$

The optimization problem becomes

$$\max_{\{u_t^{\mathcal{P}}\}} \mathbb{E}\left[\sum_{t=0}^{T} \delta^t \left\{Q(S_t) - \lambda_D D(S_t) - \lambda_K K(u_t^{\mathcal{P}}) - \lambda_R R(S_t, u_t^{\mathcal{P}})\right\}\right],$$

which is a psychological intervention problem. Therefore psychology is the nonmedical boundary case of psychiatry. $\square$

3.4 Medicalization Threshold

A psychological problem becomes psychiatric when medical authority improves expected welfare relative to nonmedical care. Define

$$\Delta W = W(A_{\mathcal{P}} \times A_{\mathcal{M}}) - W(A_{\mathcal{P}}).$$

Then psychiatric involvement is justified when

$$\Delta W > C_{\text{medicalization}},$$

where $C_{\text{medicalization}}$ includes stigma, adverse effects, overdiagnosis, opportunity cost, coercion risk, and administrative burden.

Theorem 1 (Threshold Rule)**.** *If the expected welfare gain from adding medical intervention exceeds the expected cost of medicalization, then psychiatric treatment strictly dominates purely psychological treatment.*

Proof. Let $W_{\mathcal{P}}$ denote welfare under psychological treatment alone and $W_{\mathcal{P}\mathcal{M}}$ welfare under psychological plus medical treatment. Psychiatric treatment dominates when

$$W_{\mathcal{P}\mathcal{M}} - C_{\text{medicalization}} > W_{\mathcal{P}}.$$

Rearranging gives

$$W_{\mathcal{P}\mathcal{M}} - W_{\mathcal{P}} > C_{\text{medicalization}}.$$

The left side is precisely ΔW. Hence psychiatric treatment dominates if and only if

$$\Delta W > C_{\text{medicalization}}.$$

□

4 Economics of Psychiatric Care

4.1 Mental Health as Human Capital

Let Y_i denote productivity or lifetime welfare of person i, and let M_i denote mental health capital. A reduced-form human capital equation is

$$Y_i = \alpha + \beta M_i + \gamma X_i + \varepsilon_i,$$

where X_i includes education, income, family environment, health status, and local institutions.

Mental health is economically important because it affects:

labor supply, learning, risk-taking, trust, family stability, crime, military readiness.

4.2 Cost-Benefit Model

Let C_T be treatment cost, C_U the cost of untreated illness, and B_T the benefit of treatment. Treatment is socially efficient when

$$B_T - C_T > -C_U,$$

or equivalently,

$$B_T + C_U > C_T.$$

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Table 1: Economic Channels of Psychiatric Intervention

Channel	Mechanism	Economic Implication
Labor productivity	Symptom reduction improves attendance and performance	Higher output and lower absenteeism
Education	Treatment improves concentration, sleep, and emotional regulation	Higher human capital accumulation
Family stability	Reduced distress improves caregiving and household cooperation	Lower intergenerational transmission of disadvantage
Crime and conflict	Improved impulse control and substance-use treatment	Lower enforcement and incarceration costs
Public finance	Effective treatment lowers emergency and disability costs	Reduced fiscal burden

5 Finance and Portfolio Theory of Treatment

Psychiatric care may be modeled as a portfolio of interventions. Let

$$\mathbf{w} = (w_1, w_2, \ldots, w_n)$$

be the allocation of effort, time, and money across interventions: psychotherapy, medication, sleep intervention, exercise, social support, employment assistance, and crisis planning.

Let $\boldsymbol{\mu}$ be expected benefits and Σ the covariance matrix of risks. A mean-risk treatment allocation solves

$$\max_{\mathbf{w}} \quad \mathbf{w}^\top \boldsymbol{\mu} - \frac{\rho}{2}\mathbf{w}^\top \Sigma \mathbf{w},$$

subject to

$$\sum_{j=1}^{n} w_j = 1, \qquad w_j \geq 0.$$

This formulation captures a basic clinical fact: a diversified intervention portfolio may dominate an overconcentrated one. Medication without psychological work may fail; therapy without sleep, nutrition, or medical evaluation may be underpowered; social support without risk management may be insufficient.

Table 2: Treatment Portfolio Interpretation

Intervention	Expected Return	Principal Risk
Psychotherapy	Insight, coping, behavioral change	Slow response, access cost
Medication	Symptom reduction, stabilization	Side effects, adherence issues
Sleep regulation	Mood and cognitive improvement	Implementation difficulty
Exercise	Metabolic and affective benefit	Injury, low adherence
Social intervention	Reduced isolation and stress	Dependence on local institutions
Crisis planning	Lower catastrophic risk	False reassurance if poorly designed

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6 Political Science and Institutional Authority

Psychiatry operates inside institutions: hospitals, courts, insurers, schools, armies, prisons, welfare systems, and licensing boards. This gives psychiatry a political dimension.

6.1 Authority Problem

Let A denote clinical authority and L legal authority. Psychiatric power is distinctive because in some cases

$$A \cap L \neq \varnothing.$$

That is, medical judgment may trigger legal consequences: involuntary hospitalization, disability certification, criminal responsibility assessment, fitness-for-duty decisions, or capacity evaluation.

6.2 Legitimacy Constraint

A psychiatric institution is legitimate only if

$$E[H_{\text{benefit}}] > E[H_{\text{harm}}],$$

where H_{benefit} includes reduced suffering and improved safety, while H_{harm} includes coercion, stigma, misdiagnosis, unequal access, and political misuse.

7 Geopolitics of Mental Health

Mental health systems are shaped by geopolitics through pharmaceutical supply chains, war, migration, poverty, sanctions, humanitarian crises, and international knowledge transfer.

Let country c have psychiatric capacity

$$K_c = f(D_c, P_c, T_c, G_c, S_c),$$

where D_c denotes clinician density, P_c pharmaceutical availability, T_c training infrastructure, G_c governance quality, and S_c social stability.

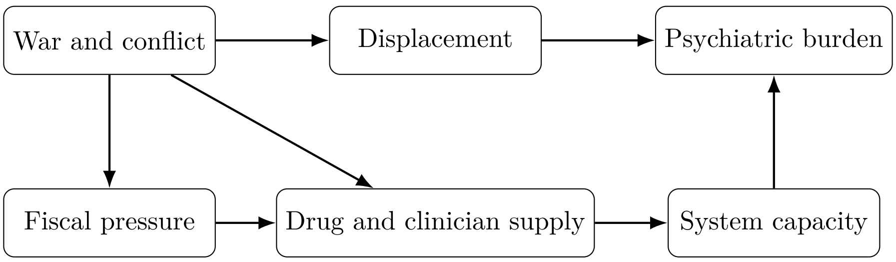

Figure 2: A geopolitical mental-health system: conflict increases burden while weakening capacity.

8 Warfare Economics and Military Psychiatry

8.1 Combat Stress as a Force-Readiness Variable

In warfare economics, mental health affects operational readiness, retention, morale, decision quality, and post-conflict social cost. Let military readiness be

$$R_t = \phi(N_t, E_t, M_t, L_t),$$

where N_t is personnel quantity, E_t equipment, M_t mental fitness, and L_t logistics.

A shock to mental fitness changes readiness by

$$\frac{\partial R_t}{\partial M_t} > 0.$$

Thus military psychiatry is not merely humanitarian; it is also an input into force preservation.

8.2 Attrition Model

Let N_t be effective force size. Suppose

$$N_{t+1} = N_t - A_t - P_t + R_t^{\text{rec}},$$

where A_t is physical attrition, P_t psychiatric attrition, and R_t^{rec} recovery or return-to-duty. Psychiatric care matters because it can reduce P_t and increase R_t^{rec}.

Let

$$P_t = \pi_0 + \pi_1 C_t + \pi_2 S_t - \pi_3 Z_t,$$

where C_t is combat intensity, S_t cumulative stress exposure, and Z_t psychiatric support capacity. If $\pi_3 > 0$, then increasing support capacity reduces psychiatric attrition.

8.3 Moral Injury and Strategic Aftermath

War generates psychological costs that survive battlefield victory. A state that wins militarily may still lose welfare if post-war psychiatric burden is large:

$$W_{\text{net}} = G_{\text{strategic}} - C_{\text{physical}} - C_{\text{economic}} - C_{\text{psychiatric}} - C_{\text{moral}}.$$

This equation clarifies why mental health is a strategic variable, not merely a clinical afterthought.

9 Statistical Model

9.1 Regression Framework

Let Y_i be symptom improvement for patient i. A standard regression model is

$$Y_i = \alpha + \tau T_i + \beta^\top X_i + \varepsilon_i,$$

where T_i indicates treatment and X_i are baseline covariates. The coefficient τ estimates treatment association, and under identification conditions may estimate a causal effect.

9.2 Causal Inference

Let $Y_i(1)$ be the potential outcome under treatment and $Y_i(0)$ under control. The average treatment effect is

$$\text{ATE} = E[Y_i(1) - Y_i(0)].$$

In randomized trials,

$$T_i \perp \{Y_i(1), Y_i(0)\},$$

so

$$\text{ATE} = E[Y_i \mid T_i = 1] - E[Y_i \mid T_i = 0].$$

In observational psychiatry, confounding is central. For example, severely ill patients may be more likely to receive medication, so naive comparisons may underestimate medication benefit.

9.3 Bayesian Updating

A psychiatrist often updates diagnostic belief as evidence arrives. Let D denote a diagnosis and E evidence. Bayes' rule gives

$$P(D \mid E) = \frac{P(E \mid D)P(D)}{P(E)}.$$

Sequentially, after evidence $E_1, \dots, E_k$,

$$P(D \mid E_1, \dots, E_k) \propto P(D) \prod_{j=1}^{k} P(E_j \mid D),$$

assuming conditional independence for simplicity.

10 Data Science and AI/ML

10.1 Predictive Risk Model

Let

$$\hat{p}_i = P(Y_i = 1 \mid X_i)$$

be a predicted probability of a clinically important event, such as relapse, hospitalization, or treatment response. A logistic model is

$$\hat{p}_i = \frac{1}{1 + \exp(-\beta^\top X_i)}.$$

A machine learning model generalizes this to

$$\hat{p}_i = f_\theta(X_i),$$

where f_θ may be a random forest, gradient boosted tree, neural network, or other function approximator.

10.2 Clinical Machine Learning Pipeline

Table 3: Algorithmic Workflow for Psychiatric Prediction

Step	Operation	Clinical Purpose
1	Define outcome Y	Avoid vague prediction targets
2	Collect features X	Include symptoms, history, context, and medical variables
3	Split data	Separate training, validation, and test sets
4	Train model f_θ	Estimate risk or treatment response
5	Calibrate predictions	Ensure predicted probabilities match observed frequencies
6	Audit fairness	Check performance across sex, age, class, ethnicity, and disability groups
7	Evaluate utility	Compare clinical benefit against false positives and false negatives
8	Monitor drift	Detect deterioration after deployment

10.3 Loss Function

For binary outcomes, cross-entropy loss is

$$\mathcal{L}(\theta) = -\sum_{i=1}^{n} \left[y_i \log f_\theta(x_i) + (1 - y_i) \log(1 - f_\theta(x_i)) \right].$$

However, clinical prediction requires more than statistical accuracy. The model must also satisfy calibration, interpretability, privacy, robustness, and ethical constraints.

11 Decision Theory

Let the psychiatrist choose between no medication (a_0) and medication (a_1). Let B be expected benefit and H expected harm. Choose medication when

$$E[B \mid a_1] - E[H \mid a_1] > E[B \mid a_0] - E[H \mid a_0].$$

Equivalently,

$$\Delta B > \Delta H.$$

This simple inequality captures the clinical logic of medical addition: medicine is justified only when its expected marginal benefit exceeds its expected marginal harm.

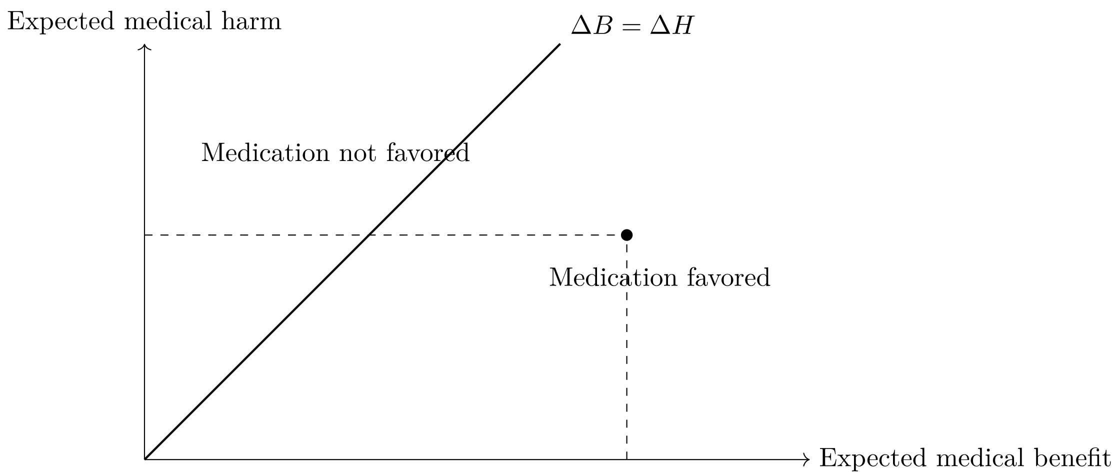

Figure 3: A decision threshold for adding medical intervention to psychological care.

12 Ethics

The phrase "psychology with medicine" must be ethically constrained. Medicine adds power. Power creates risk. Therefore psychiatry requires special attention to autonomy, informed consent, proportionality, nonmaleficence, beneficence, justice, and epistemic humility.

12.1 Four Ethical Constraints

Autonomy:	respect patient agency where possible,
Beneficence:	act for patient welfare,
Nonmaleficence:	avoid unnecessary harm,
Justice:	avoid discriminatory allocation or coercion.

12.2 Coercion Constraint

Let C denote coercive intervention. Coercion is justified only if

$$E[H_{\text{prevented}} \mid C] - E[H_{\text{caused}} \mid C] > \kappa,$$

where κ is a high ethical threshold. This formalizes the principle that coercion must be exceptional, proportionate, reviewable, and justified by serious risk.

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13 Comparison with Adjacent Disciplines

Table 4: Psychiatry Compared with Adjacent Fields

Field	Primary Object	Distinctive Instrument
Psychology	Mind and behavior	Testing, therapy, behavioral science
Psychiatry	Mind, behavior, and medical risk	Diagnosis, medication, risk triage, medical authority
Neurology	Nervous system disease	Lesion localization, neurophysiology, neurological examination
Social work	Person-in-environment	Case management, welfare access, family and community systems
Public health	Population risk	Prevention, surveillance, policy, resource allocation
Economics	Scarcity and incentives	Cost-benefit analysis, optimization, institutional design
Political science	Power and governance	Legitimacy, authority, rights, bureaucracy

14 An Integrated Model

The central model of this paper can be summarized as

$$\text{Psychiatry} = \text{Psychology} + \text{Medicine} + \text{Risk Authority} + \text{Institutional Responsibility}.$$

More explicitly,

$$\Psi_{\text{psychiatry}} = \Psi_{\text{psychology}} \oplus \mathcal{M}_{\text{medicine}} \oplus \mathcal{R}_{\text{risk}} \oplus \mathcal{I}_{\text{institution}},$$

where $\oplus$ denotes structured integration rather than simple addition.

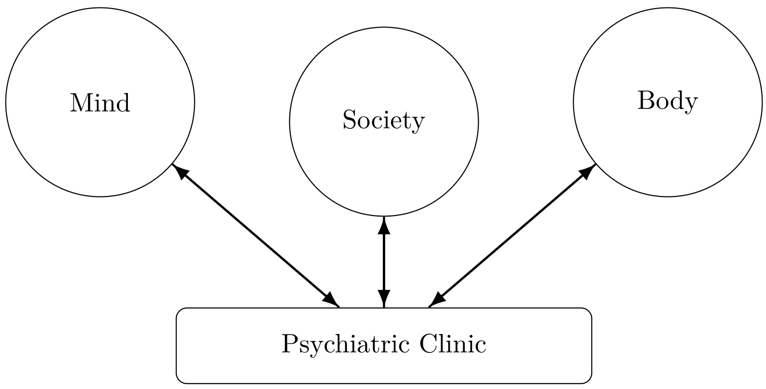

Figure 4: Psychiatry as a feedback system linking mind, body, and society.

15 Policy Implications

The thesis implies five policy conclusions.

1. Mental health systems should integrate psychotherapy, medication, primary care, social care, and crisis response.
2. Psychiatric training should include psychology, neuroscience, pharmacology, statistics, ethics, economics, and law.
3. Public funding should evaluate both clinical outcomes and social spillovers.
4. AI systems in psychiatry must be clinically validated, calibrated, audited, and governed.
5. Military and humanitarian psychiatry should be treated as strategic public-health infrastructure, not merely post-crisis charity.

16 Limitations

The phrase "psychology with medicine" is analytically useful but incomplete. Psychiatry is also shaped by anthropology, sociology, law, economics, philosophy, neuroscience, and politics. Moreover, many psychiatric conditions cannot be cleanly separated into psychological and biological components. The phrase should therefore be read as a structural approximation: psychiatry inherits the psychological object of care and adds the medical apparatus of diagnosis, intervention, and risk responsibility.

17 Conclusion

Psychiatry is neither merely psychology nor merely medicine. It is the discipline that treats psychological suffering under medical uncertainty. Its clinical grammar is psychological: symptoms appear as mood, thought, fear, impulse, memory, behavior, identity, relationship, and meaning. Its institutional grammar is medical: diagnosis, pharmacology, risk, consent, capacity, hospitalization, public health, and duty of care.

Thus the formula

$$\boxed{\text{Psychiatry} = \text{Psychology with Medicine}}$$

is not a reduction. It is a synthesis. Psychiatry is psychology when the mind is the object; it is medicine when the body, risk, and treatment authority become indispensable; and it is political economy when mental health becomes a matter of institutions, war, welfare, productivity, justice, and state capacity.

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Glossary

Psychiatry
A medical discipline concerned with mental disorders, psychological suffering, behavioral disturbance, and medical risk.

Psychology
The scientific study of mind, behavior, cognition, emotion, learning, personality, and social relation.

Medicalization
The process by which a condition becomes interpreted and managed through medical categories and institutions.

Biopsychosocial Model
A framework in which biological, psychological, and social factors jointly determine health and illness.

Pharmacology
The study of drugs, including mechanism, dose, efficacy, side effects, and interaction.

Causal Inference
The statistical study of cause-and-effect relations, especially through potential outcomes, experiments, and quasi-experiments.

Average Treatment Effect
The expected difference between outcomes under treatment and outcomes under control.

Calibration
Agreement between predicted probabilities and observed frequencies.

Moral Injury
Psychological and ethical distress arising from violation of deeply held moral commitments, often discussed in war and trauma contexts.

Warfare Economics
The study of resource allocation, incentives, production, destruction, logistics, and human capital under conditions of war.

Psychiatric Attrition
Loss of effective personnel or functioning due to mental-health deterioration.

Institutional Legitimacy
The justified authority of institutions based on fairness, effectiveness, accountability, and rights protection.

The End